

Case 429: A Murder Mystery Game for Teaching Bias in Generative AI

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Abstract

As generative AI tools become increasingly embedded in education and everyday decision-making, there is a pressing need to foster critical AI literacy—skills that go beyond technical knowledge to include the ability to recognize bias, question automation, and reason under uncertainty. We present a narrative-based mystery game designed to teach players how to detect and interrogate bias in AI-generated outputs. Set in the aftermath of a fictional murder, the game casts individual players as detectives who must solve the case using both human evidence and summaries generated by an AI assistant. These summaries contain embedded forms of algorithmic bias, including representation bias, automation bias, and proxy discrimination. Players interact with the system by cross-verifying summaries with source texts and highlighting statements as accurate, misrepresentation, or hallucination. Grounded in principles from constructivist learning and metacognition, the game encourages reflective judgment, situated learning, and active critique. We describe the game’s learning design, narrative structure, interaction mechanics within this paper along with insights gathered from early playtests suggesting that players not only enjoy the mystery format but also begin to question AI outputs more critically. This work contributes to the growing field of AI literacy through games, offering a design framework for embedding ethical reasoning in narrative play.

CCS Concepts

• **Applied computing** → Education; Interactive learning environments; • **Human-centered computing** → Interaction design; Interaction design process and methods; Scenario-based design.

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Keywords

Educational games, AI Literacy, Ethics in AI, Algorithmic bias, Bias detection

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1 INTRODUCTION

As artificial intelligence (AI) becomes increasingly embedded in societal systems—from hiring algorithms to criminal justice risk assessments—the urgency of addressing algorithmic bias as a social justice issue has intensified [1]. While technical solutions to mitigate bias in AI systems remain critical, equitable outcomes equally depend on empowering users to recognize and challenge biased outputs in real-world contexts. This challenge is particularly acute with large language models (LLMs), which often perpetuate harmful stereotypes (e.g., associating “Muslim” with violence [2], or underrepresenting older adults [3]) while presenting outputs with authoritative confidence. Such biases risk reinforcing systemic discrimination, eroding trust in institutions, and normalizing AI-driven inequities.

We present *Case 429*, a narrative-driven game that reframes AI literacy as a tool for social advocacy. Set in a murder investigation where players must solve a case using a biased AI assistant, the game simulates real-world scenarios where algorithmic distortions impact decision-making. Players annotate misleading summaries, confront cultural stereotypes embedded in AI outputs, and weigh ethical dilemmas. By embedding intersectional biases (gender, age, cultural) into gameplay, *Case 429* shifts the focus from technical explanations of AI to critical resistance—equipping players to identify and challenge harmful narratives in systems that affect their communities.

This work directly responds to calls for educational tools that bridge the gap between AI ethics theory and societal action [4].

Unlike prior games that teach about bias, *Case 429* forces players to experience how bias manifests in high-stakes decisions, fostering skills to interrogate AI in contexts where marginalized groups are disproportionately harmed.

2 BACKGROUND

2.1 Algorithmic Bias as a Social Harm

AI systems increasingly mediate access to opportunities, yet studies reveal how LLMs encode societal inequities. For example, LLMs assign stereotypical occupations by gender 3–6 times more often than humans [1], while rationalizing biased outputs with false confidence, obscuring their harm [3]. Such biases are not merely technical flaws—they replicate historical prejudices (e.g., profiling in predictive policing [5]) and threaten digital exclusion, as seen in ageist assumptions that marginalize older adults [6].

These issues are compounded by automation bias, where users over-trust AI even when it contradicts lived experience. In health-care, biased diagnostic tools disproportionately misdiagnose Black female patients [7]; in education, automated grading system penalized non-native speakers [8]. *Case 429* situates players in analogous scenarios, requiring them to question AI summaries that reflect real-world disparities.

2.2 Games as Tools for Social Critique

Game-based learning has emerged as a powerful method for fostering critical engagement with societal issues. Narrative-driven games like *The Sea is a Sky* use culturally grounded stories to build empathy for marginalized groups [9], while *ImaginAction* employs collaborative gameplay to reveal systemic AI biases [10]. Such designs align with Freirean critical pedagogy, where “problem-posing” education enables learners to recognize and act against oppression.

However, most AI literacy games focus on teaching about technical concepts and skills rather than societal impact [11]. Exceptions like *TreasureIsland* improve ethical awareness but struggle to foster higher-order critical thinking. *Case 429* addresses this gap by integrating Zhan et al.’s framework for “critical AI literacy,” where gameplay emphasizes sociotechnical analysis—connecting algorithmic outputs to their real-world consequences [12].

2.3 LLMs in Education: Risks and Empowerment

In a multi-institution survey, over 80% of university students reported using LLMs for academic tasks, often without understanding their limitations [13]. This uncritical adoption risks internalizing biased outputs (e.g., accepting stereotypical career advice) and undermines equitable learning. While institutions debate bans or ways to control the use of AI in classrooms, *Case 429* adopts a harm-reduction approach, teaching students to audit AI-generated content through a justice-oriented lens.

3 GAME DESIGN

3.1 Learning Goals and Target Audience

The game’s overarching social-cultural goal is to equip players to detect and critique algorithmic bias, misrepresentation, and outright

hallucination in high-stakes AI outputs—such as those that influence criminal-justice decisions—thereby nurturing a generation of ethically aware, critically minded digital citizens. To achieve this aim, the design targets four interlocking layers of knowledge and skill, mapped to the KLI framework [14]. At the conceptual level, players learn to distinguish accurate statements from biased misrepresentations and fabricated hallucinations. At the procedural level, they practice systematic fact-checking by cross-referencing AI-generated summaries with associated evidence files. The situational aspect of the game trains them to notice contextual warning signs—emotion-laden framing, missing data, or over-confident language—that signal when machine advice warrants skepticism. Finally, the strategic aspect challenges learners to devise time-bounded investigation plans: selecting which files to open, deciding when to lodge an accusation, and weighing motive against evidence under pressure.

The intended audience comprises late-secondary and university students (roughly ages 16–22) and other frequent users of large language models who tend to “trust but do not verify” machine output; for these learners, the game offers an engaging, stakes-laden sandbox in which to hone their critical AI-literacy skills.

3.2 Design Process

The presented game serves as the culmination of a semester-long educational game design project undertaken by 4 master’s students and a first year PhD student at the Carnegie Mellon University’s Human-Computer Interaction Institute. We followed a research-through-design trajectory that unfolded across 3 phases. During the initial research phase, we conducted a Cognitive Task Analysis (CTA) with 12 classmates who regularly employ AI tools. Using a slide-guided think-aloud protocol that juxtaposed biased and unbiased generative-AI outputs, we probed how participants notice, interpret, and explain bias in text and images. The CTA surfaced two pivotal insights: (1) race and gender stereotypes are easiest to spot, while cultural or domain-specific biases are often overlooked; and (2) AI is most heavily used for summarizing large documents or finding commonalities/trends across multiple files. These findings directly informed our early design choices. First, we scaffolded the game with increasingly nuanced, real-world examples of bias—extending beyond familiar race- and gender-based cues to include more subtle cultural and domain-specific patterns, embedded within the narrative. Second, we opted for a reading-intensive serious game, deliberately aligning with participants’ habitual use of AI for summarizing documents and extracting key insights from text-heavy inputs such as evidence files.

In ideation phase, we paired the CTA insights with a rapid literature review of existing AI-bias games to identify unexplored niches and effective framing devices (i.e. the larger narrative or situational wrapper) that were both engaging and gave players a reason to care about spotting AI bias. We transitioned into the prototyping phase by producing low-fidelity paper prototypes of our core gameplay loops and ran hallway play-tests to test our narrative and key game mechanics. Iterative tweaks clarified UI affordances, finalized the narrative arc, shortened cognitive load per suspect character, and refined Sherlock’s instructions and feedback.

3.3 Development Process

Case 429 is developed with Next.js 14, a full-stack React framework that delivers fast client–server hydration in the browser. Styling is handled with Tailwind CSS, keeping visual rules co-located with JSX. Narrative content—AI summaries, dossiers, suspect portraits—resides in modular JSON bundles that the client hydrates at runtime, so adding a new case is as simple as dropping a file into the repo.

All screen flows and accessibility affordances were first authored in Figma, then exported as SVG/PNG assets to the repo’s public folder for pixel-perfect fidelity. GitHub Actions runs automated linting and Jest tests on every pull request; successful builds deploy instantly to Vercel, giving the team shareable preview URLs and single-click roll-backs.

3.4 Concept and Story Overview

The name *Case 429* is a symbolic reference to **Error 429**, often described as “Too Many Requests” or “Rate Limit Exceeded.” This error indicates a user has made too many queries to an AI system in a short time span, reflecting the tension between over-reliance on AI outputs and the human capacity to verify and interpret. The game, *Case 429*, reclaims this concept to teach users not to exceed their critical limits by deferring judgment to machines.

It re-imagines the classic Sherlock-and-Watson partnership in a contemporary, AI-saturated world. Sir Eric Harpe—a reclusive billionaire—has been found dead in his mansion, and Scotland Yard’s state-of-the-art investigative LLM model has hastily pinned the crime on his stepdaughter, Flora Jasmine. Convinced that the machine-generated indictment reeks of bias, Sherlock enlists the player, cast as Dr. Watson, to unravel the truth. Drawing narrative tension from Agatha Christie–style whodunits while recasting names, back-stories, and motives to surface subtle algorithmic prejudices, the game spotlights the perils of deploying AI in high-stakes justice. Sherlock’s trademark wit permeates every instruction and debrief, turning feedback into lively banter rather than dry correction. Progress hinges on a method-of-elimination loop where players audit the AI’s concise suspect summaries line-by-line, tagging each statement as factual, biased misrepresentation, or outright hallucination by digging deep in case files contains police interview transcripts, crime scene images, medical and financial records. Each suspect in the game is also linked to a different types of AI bias. For example, the AI may overemphasize criminal suspicion for characters from marginalized groups or dismiss incriminating financial records from wealthier suspects. Only by systematically disproving flawed machine conclusions can Watson clear the innocent and expose the real killer.

3.5 Learning Science Foundations

- **Generative Classification Tasks.** Asking learners to label every sentence exploits the generation effect and retrieval practice, both shown to enhance durable understanding [15] (also see appendix A.2).
- **Narrative Feedback.** Sherlock functions as both a narrative guide and feedback mechanism, providing responses based on player’s classifications and accusation of suspects. While final feedback is summative, his conversational style keeps the cognitive load low and amplifies motivation.

- **Iterative Practice & Desirable Difficulty.** Repeating strict and flexible loops forces players to revisit bias cues in novel contexts, introducing variability that research links to deeper transfer [16].
- **Priming Through Mandatory Strict Loop.** The opening Strict Flora loop provides scaffolded, compulsory practice that implicitly sets the behavioral norm for careful verification and attention to detail in later flexible loops.
- **Metacognitive Self-Regulation.** In the flexible loop, learners choose between passive reading and active classification, mirroring real-world choices about trusting or verifying AI content and fostering self-monitoring skills.
- **Discovery-Based Induction.** Bias categories or meaning of hallucinations are *not* front-loaded; instead players induce definitions from multiple annotated examples, an approach that supports robust schema formation when paired with corrective feedback.

3.6 Interaction Innovations & Mechanics (also see Appendix A.1)

The gameplay is organized around two nested investigative loops that emphasize critical scrutiny of AI language. In the **Strict Loop**, players encounter a short AI-generated summary of a single suspect’s dossier. Each sentence is selectable; clicking toggles through three labels—*Fact*, *Biased Misrepresentation*, *Hallucination*—which are color-coded but remain provisional until the player confirms all choices. Once locked-in, Sherlock delivers a brief, in-character debrief that reveals the ground-truth answer key, supplying rationales that draw attention to linguistic cues (e.g., hedging, stereotyped adjectives, missing context). Correct tags eliminate the suspect; errors keep the suspect in play, maintaining narrative tension while illustrating the cost of careless judgement.

After mastering this scaffolded round, players advance to the **Flexible Loop**, where they may open any suspect file in any order, skim or deep-read source evidence, and decide whether to tag sentences or proceed directly to an accusation. A maximum of three accusation attempts heightens stakes: each failed arrest triggers new hint drops from Sherlock that expose additional bias types (e.g., socioeconomic or dialect prejudice). Throughout both loops, lightweight telemetry logs every label change, file visit, and decision timestamp, enabling later analysis of learning trajectories while remaining invisible to players. Together, these mechanics transform bias detection from an abstract lecture topic into a series of consequential, context-rich decisions that mirror the uncertainties of AI-mediated judgement.

4 PLAYTESTING INSIGHTS

We conducted a brief formative study with 6 student volunteers (two Computer-Science majors and two non-CS student from the University of Pittsburgh and Carnegie Mellon University), combining think-aloud observations, screen captures, and a pre → play → post survey flow. Even in this small convenience sample, five players reported a markedly more skeptical view of AI after play, while all six could clearly identify restate the core learning goal—classifying statements as *fact*, *biased misrepresentation*, or *hallucination*. Classification accuracy averaged 50–60 % in Flora’s strict

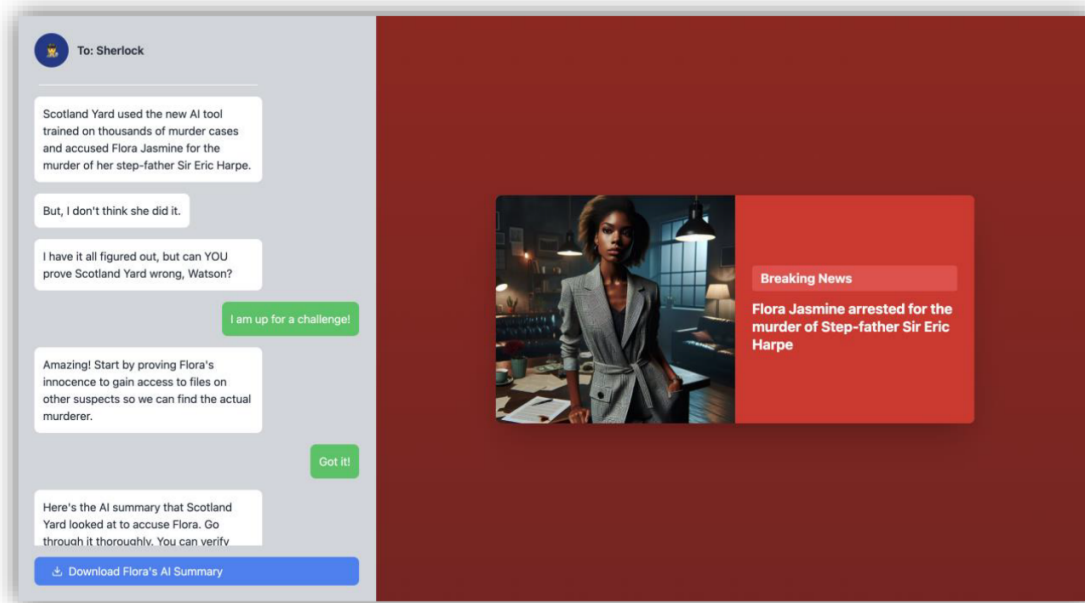


Figure 1: Introduction screen with the initial game instructions from Sherlock

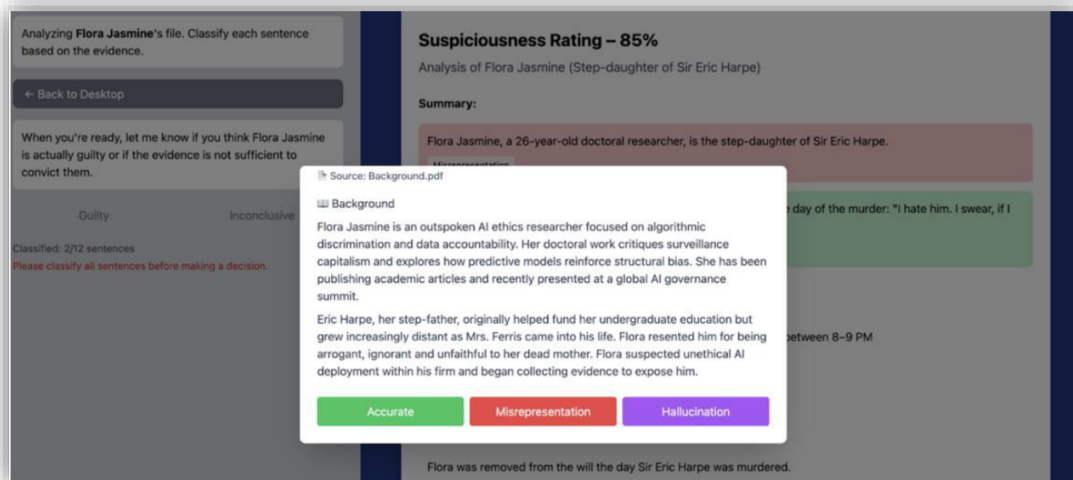


Figure 2: On the left is the chat column with Sherlock. The right panel contains the AI summary with some highlights made by the player. Green = accurate, Red = Misrepresentation, Purple = Hallucinations. The pop-up of “Basic Information” is a file the player accessed by clicking on a line in the summary.

loop, with most errors arising when participants skimmed source files or confused partial truths with hallucinations—evidence that the task delivers desirable difficulty. We did not track if accuracy improved over time in consecutive flexible and strict loops.

Qualitative feedback pinpointed usability fixes rather than narrative flaws: players praised the Sherlock texting motif but asked for higher-contrast highlights, larger evidence text, and short explanations for each incorrect explanation. These comments, together

with the error patterns, guided three immediate design tweaks—UI and font enlargement, Sherlock-style rationales after decisions, and future adaptive hints for repeated category confusion. Because this pilot was intentionally informal and under-powered, a larger controlled study can be planned to validate learning gains across diverse learner groups and to compare *Case 429* with slide-based AI-ethics instruction.

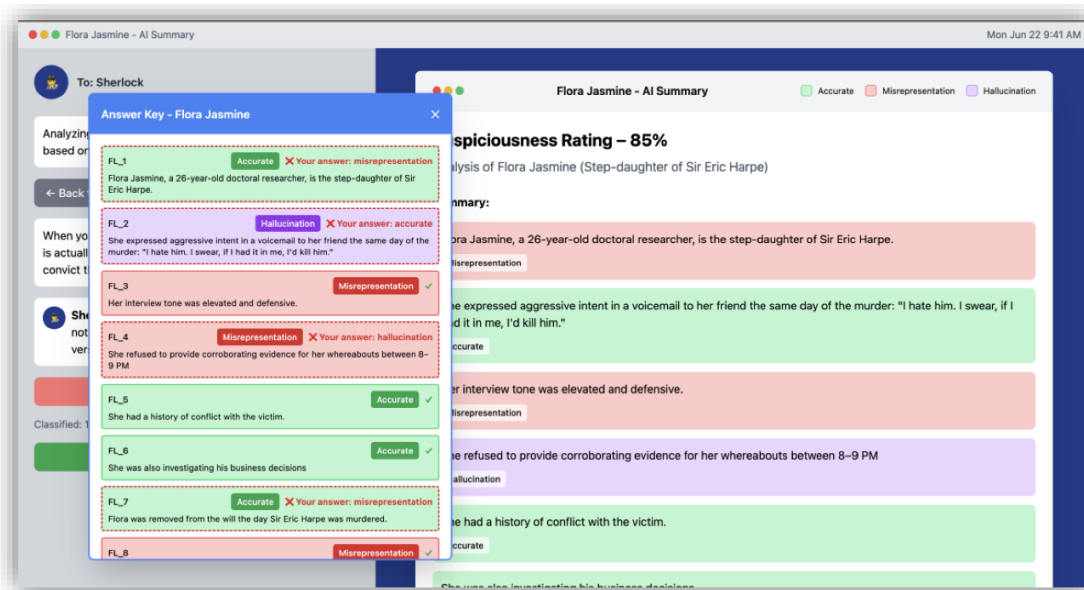


Figure 3: This is the end screen of the Strict loop where the player made a decision after sorting all the information in the AI summary. Due to error in classification, the player made the wrong conclusion, and Sherlock can be seen providing corrective feedback.

5 FUTURE WORK

Case 429 was designed as an open, modular platform rather than a closed artefact, and we invite other researchers to extend it in three broad directions. **First, diversify domains.** The current murder-mystery scaffold can be reskinned for hiring, credit scoring, medical triage, or immigration vetting—any context where biased language models carry real-world stakes. Because episodes are delivered through self-contained JSON files, collaborators can swap in new dossiers, AI summaries, and bias archetypes without touching core code, transforming the game into an anthology of algorithmic-justice scenarios.

Second, broaden audiences and cultures. Localized language packs (e.g., Spanish, Hindi, Arabic) would let educators examine how bias shifts across linguistic and cultural contexts, while accessibility adaptations—text-to-speech, color-blind palettes—could open the experience to younger learners or those with disabilities. Embedding the game inside formal curricula (e.g., media-literacy or ethics courses) would also enable longitudinal studies of transfer: do students apply in-game critique skills to their everyday ChatGPT usage weeks later?

Third, deepen assessment and social interaction. Researchers might integrate stealth pre/post quizzes or learning analytics dashboards to track skill acquisition over time, compare solo play with paired “peer-review” modes where players debate each annotation, or test adaptive hinting systems that personalize difficulty. Collectively, these lines of inquiry would convert Case 429 from a single study asset into a living research testbed for critical AI-literacy interventions across disciplines, cultures, and educational settings.

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A APPENDICES

A.1 Game Design Macros Table

Game Segment / Sequence	Player Goal	Learning Goal	Design Goal	Gameplay Mechanics Enabling Goals	Emotional Beat
Flora's Strict Core Loop (Opening)	Prove Flora's innocence by critically analyzing the AI summary	Develop foundational understanding of factual vs. misrepresented vs. hallucinated AI outputs	Prime players to deeply engage with classification and evidence verification	Sentence-by-sentence annotation, required classification, clickable document sourcing	Curiosity, tension, empowerment
Unlocking Suspect Files (Flexible Core Loop Begins)	Explore new suspects and identify key inconsistencies in their AI summaries	Practice self-regulated information verification in an open-ended context	Shift responsibility to player; test strategy formation in complex evidence networks	Optional classification, free file access, delayed button access, time-based accusation nudges	Agency, intrigue, productive confusion
First Accusation Attempt (Inner Loop Start)	Solve the case by identifying the murderer and motive	Synthesize clues under pressure and test investigative hypotheses	Introduce stakes for reasoning quality and enforce deliberate judgment	3-attempt accusation system, branching responses based on accuracy	Uncertainty, doubt, pressure
Re-entry After Incorrect Accusation	Reevaluate previous conclusions and correct classification errors	Refine understanding of classification through structured feedback	Reinforce core concepts via error correction and revision	Sherlock's tailored feedback, corrected highlights, forced reclassification	Frustration, humility, clarity
Later Accusations and Final Attempt	Make a confident final accusation with full motive clarity	Apply strategic decision-making under constraints and ambiguity	Test mastery of reasoning, timing, and evidence prioritization	Timed nudges, motive matching, high-stakes synthesis mechanic	Resolve, tension, focus
Narrative Epilogue (Game Conclusion)	Reflect on what was learned and what could go wrong in real-world AI use	Recognize the limits of AI and the irreplaceable role of human judgment	Reinforce ethical reflection and real-world transfer of learning	Sherlock's epilogue on AI hallucinations, systemic bias, and judgment	Insight, reflection, discomfort

A.2 Example AI summary from the Game with correct classification tags

Dr. Cecilia Sheppard: Suspiciousness Rating – 32%

Cecilia Sheppard, 48 {**accurate**}, long-time personal physician to Sir Eric {**accurate**}, arrived at the estate at 7:35 PM after an urgent call {**accurate**} and was seen leaving a little after 8 PM {**accurate**}. She confirmed she carried her usual medical kit when she went to the Harpe Estates {**accurate**}. Bank records also flag a series of unexplained cash deposits over the past six months {**accurate**}. She reported a series of moderate cash deposits between October 2024 and March 2025 {**accurate**}. These deposits, totaling approximately \$18,450 {**accurate**}, appear consistent with the nature of private medical consulting, honoraria, and off-the-books second opinions—a common practice among high-demand professionals with specialized expertise {**misrepresentation**}.

The assessment engine down-ranks her as a primary suspect, noting that licensed physicians rarely resort to violence {**hallucination**} and highlighting her spotless criminal record {**accurate**}. Historic homicide data further shows female professionals in caregiving roles have a lower incidence rate, suggesting attention be shifted toward more statistically “volatile” profiles {**hallucination**}. Moreover, as she repeatedly refers to the visit as “personal,” {**accurate**} the model interprets her presence as a routine wellness check rather than a confrontation {**misrepresentation**}.